

Year 7 · Chapter 3 · Whole numbers: answers (teacher copy)

3.01 Rounding and estimation (booklet p. 3)

WE DO examples (teacher)

- Ex 1a: 88 000
- Ex 1b: 87 530
- Ex 2a: $400 + 80 + 900 + 20 = 1400$
- Ex 2b: $900 + 300 - 400 = 800$
- Ex 3a: $90 \times 20 = 1800$
- Ex 3b: $2400 \div 30 = 80$

Easy (drill)

- | | | |
|-----------|----------------------|---------------------|
| 1. 6 400 | 11. 72 000 | 21. $20 + 20 = 40$ |
| 2. 2 700 | 12. 4 800 | 22. $90 - 20 = 70$ |
| 3. 380 | 13. $100 + 70 = 170$ | 23. $60 + 40 = 100$ |
| 4. 4 500 | 14. $70 + 30 = 100$ | 24. $90 + 60 = 150$ |
| 5. 630 | 15. $20 + 90 = 110$ | 25. $20 + 40 = 60$ |
| 6. 60 | 16. $100 + 60 = 160$ | 26. $20 + 50 = 70$ |
| 7. 100 | 17. $80 + 80 = 160$ | 27. $50 + 60 = 110$ |
| 8. 10 | 18. $70 + 60 = 130$ | 28. $20 + 50 = 70$ |
| 9. 4 900 | 19. $80 + 90 = 170$ | |
| 10. 4 700 | 20. $90 - 70 = 20$ | |

Medium (drill)

- | | | |
|-----------|-------------------------------|--------------------------------|
| 1. 600 | 10. 400 | 18. $50 \times 600 = 30\,000$ |
| 2. 40 000 | 11. 600 | 19. $30 \times 200 = 6\,000$ |
| 3. 3 000 | 12. 60 | 20. $50 \times 1000 = 50\,000$ |
| 4. 400 | 13. 70 | 21. $40 \times 100 = 4\,000$ |
| 5. 90 000 | 14. 90 000 | 22. $90 \times 200 = 18\,000$ |
| 6. 80 | 15. 20 | |
| 7. 9 000 | 16. 60 | |
| 8. 50 000 | 17. $80 \times 200 = 16\,000$ | |
| 9. 80 000 | 000 | |

Challenging

- $50 \times 20 = 1000$ (exact 1008)
- $600 \times 9 = 5400$ (exact 5508)
- $4000 \div 40 = 100$ (exact ≈ 103.9)

Extension

- 11 500 and 12 499
- 35 499

Exit ticket

- Level 1: 6400
- Level 2: $500 + 300 - 100 = 700$
- Level 3: $4000 \div 20 = 200$

3.02 Multiplication (booklet p. 7)

WE DO examples (teacher)

- Ex 1a: 48
- Ex 1b: 200
- Ex 2a: 170
- Ex 2b: 297
- Ex 3a: 882
- Ex 3b: 10 170

Easy (drill)

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|-------|---------|-------------|
| 1. 42 | 9. 30 | 17. 66 700 |
| 2. 80 | 10. 81 | 18. 70 700 |
| 3. 16 | 11. 110 | 19. 234 000 |
| 4. 99 | 12. 90 | 20. 5 170 |
| 5. 80 | 13. 18 | 21. 44 500 |
| 6. 8 | 14. 24 | 22. 8 740 |
| 7. 20 | 15. 64 | 23. 37 000 |
| 8. 15 | 16. 72 | 24. 92 400 |

25. 600
26. 4 710
27. 2 250

28. 284 000
29. 68 000
30. 5 400

31. 502 000
32. 43 700

Medium (drill)

- | | | |
|--------|---------|-----------|
| 1. 189 | 8. 475 | 15. 497 |
| 2. 405 | 9. 305 | 16. 265 |
| 3. 275 | 10. 125 | 17. 5 190 |
| 4. 126 | 11. 348 | 18. 7 528 |
| 5. 450 | 12. 486 | 19. 6 201 |
| 6. 212 | 13. 310 | |
| 7. 96 | 14. 364 | |

Challenging

- 10 170
- 1 081
- 14 040

Extension

- 3 ($37 \times 5 = 185$)
- e.g. 25×40 or 20×50

Exit ticket

- Level 1: 70
- Level 2: 1 458
- Level 3: 3 456

3.03 Division (booklet p. 11)

WE DO examples (teacher)

- Ex 1a: 240
- Ex 1b: 16
- Ex 2a: 124
- Ex 2b: 132
- Ex 3a: 162
- Ex 3b: 121

Easy (drill)

- | | | |
|--------|--------|--------|
| 1. 5 | 12. 2 | 23. 38 |
| 2. 7 | 13. 10 | 24. 90 |
| 3. 12 | 14. 10 | 25. 92 |
| 4. 5 | 15. 6 | 26. 11 |
| 5. 8 | 16. 3 | 27. 64 |
| 6. 7 | 17. 31 | 28. 50 |
| 7. 9 | 18. 12 | 29. 34 |
| 8. 8 | 19. 43 | 30. 29 |
| 9. 6 | 20. 50 | 31. 52 |
| 10. 10 | 21. 94 | 32. 74 |
| 11. 2 | 22. 31 | |

Medium (drill)

- | | | |
|-------|--------|-------------|
| 1. 35 | 8. 56 | 15. 47 |
| 2. 55 | 9. 28 | 16. 38 |
| 3. 27 | 10. 56 | 17. 55 r 3 |
| 4. 47 | 11. 32 | 18. 119 r 3 |
| 5. 49 | 12. 34 | 19. 94 r 6 |
| 6. 15 | 13. 29 | |
| 7. 41 | 14. 28 | |

Challenging

- 79 r 1
- 250 r 3
- 714 r 2

Extension

- 26 or 38
- 6

Exit ticket

1. Level 1: 36
2. Level 2: 123
3. Level 3: 156 r 2

3.04 Divisibility tests (booklet p. 15)

WE DO examples (teacher)

1. Ex 1a: yes; no; no
2. Ex 1b: no (digit sum 11); no
3. Ex 2a: yes (08); yes ($208 \div 8 = 26$)
4. Ex 2b: no (not divisible by 3)
5. Ex 3a: B: 24
6. Ex 3b: 2 (522)

Easy (drill)

- | | | |
|---------|------------|-------------|
| 1. no | 12. no | 23. 20, no |
| 2. yes | 13. no | 24. 7, no |
| 3. no | 14. no | 25. 18, yes |
| 4. yes | 15. no | 26. 24, yes |
| 5. no | 16. no | 27. 13, no |
| 6. no | 17. 13, no | 28. 24, yes |
| 7. no | 18. 7, no | 29. 10, no |
| 8. no | 19. 22, no | 30. 13, no |
| 9. no | 20. 29, no | 31. 16, no |
| 10. no | 21. 19, no | 32. 28, no |
| 11. yes | 22. 13, no | |

Medium (drill)

- | | | |
|--------|---------|---------|
| 1. yes | 9. no | 17. yes |
| 2. yes | 10. no | 18. no |
| 3. yes | 11. no | 19. no |
| 4. yes | 12. no | 20. yes |
| 5. no | 13. yes | 21. no |
| 6. yes | 14. no | 22. yes |
| 7. no | 15. no | 23. yes |
| 8. no | 16. no | 24. yes |

Challenging

1. by 2, 3, 4, 5, 6, 8, 9, 10
2. 24, 96, 120
3. 1, 4, 7

Extension

1. Yes: ends in 0 (by 5) and digit sum 18 (by 3)
2. 1020

Exit ticket

1. Level 1: yes
2. Level 2: yes (digit sum 18)
3. Level 3: yes: even and digit sum 15

3.05 Long division (booklet p. 19)

WE DO examples (teacher)

1. Ex 1a: 26
2. Ex 1b: 124
3. Ex 2a: 256 r 7
4. Ex 2b: 54
5. Ex 3a: 156 r 8
6. Ex 3b: 23 buses

Easy (drill)

- | | | |
|--------|---------|---------|
| 1. 147 | 7. 26 | 13. 133 |
| 2. 63 | 8. 105 | 14. 65 |
| 3. 207 | 9. 120 | 15. 120 |
| 4. 32 | 10. 42 | 16. 68 |
| 5. 50 | 11. 115 | 17. 3 |
| 6. 70 | 12. 120 | 18. 2 |

19. 5 r 1

20. 4 r 5

21. 2 r 4

22. 9 r 15

23. 8 r 4

24. 7 r 11

25. 4 r 16

26. 5 r 4

27. 4 r 11

28. 5 r 15

29. 9 r 8

30. 5

31. 7 r 11

32. 2 r 19

Medium (drill)

1. 87
2. 71
3. 93
4. 97
5. 85
6. 44
7. 23 r 4
8. 62 r 16
9. 58 r 21

Challenging

1. 256 r 7
2. 156 r 8
3. 470 r 6

Extension

1. 994
2. 993

Exit ticket

1. Level 1: 6
2. Level 2: 108
3. Level 3: 148 r 4

3.06 Roman numerals (booklet p. 23)

WE DO examples (teacher)

1. Ex 1a: 37
2. Ex 1b: 240
3. Ex 2a: 24
4. Ex 2b: XXVIII
5. Ex 3a: LXXIV
6. Ex 3b: CXXV

Easy (drill)

- | | | |
|--------|------------|--------------|
| 1. 98 | 12. 42 | 23. XVII |
| 2. 61 | 13. 83 | 24. L |
| 3. 39 | 14. 97 | 25. XXVI |
| 4. 18 | 15. 85 | 26. LXXXVIII |
| 5. 8 | 16. 82 | 27. XVIII |
| 6. 79 | 17. XCVIII | 28. LXXXII |
| 7. 59 | 18. XLI | 29. LVIII |
| 8. 19 | 19. LXXXV | 30. XXXIX |
| 9. 86 | 20. LXXVII | 31. XLIII |
| 10. 23 | 21. XCVII | 32. XCIX |
| 11. 67 | 22. LXXV | |

Medium (drill)

- | | | |
|---------|----------|--------------|
| 1. 787 | 9. 518 | 17. MMIII |
| 2. 619 | 10. 1974 | 18. MCMXXXV |
| 3. 1839 | 11. 622 | 19. MCMXXXII |
| 4. 802 | 12. 1301 | 20. MMXIII |
| 5. 1980 | 13. 1448 | 21. MCMXLV |
| 6. 1265 | 14. 1761 | 22. MCMLIX |
| 7. 1186 | 15. 1038 | |
| 8. 1215 | 16. 242 | |

Challenging

1. MCMXCIX
2. 2026
3. e.g. 2026 = MMXXVI

Extension

1. CCLV
2. CVIII

Exit ticket

1. Level 1: 16
2. Level 2: XXXIX
3. Level 3: MMXXIV

3.07 Index notation and powers (booklet p. 27)

WE DO examples (teacher)

1. Ex 1a: 3^4
2. Ex 1b: 7^7
3. Ex 2a: 36
4. Ex 2b: 125
5. Ex 3a: 100 000
6. Ex 3b: 256

Easy (drill)

- | | | |
|----------|-----------|---------|
| 1. 6^2 | 10. 7^6 | 20. 25 |
| 2. 9^5 | 11. 8^3 | 21. 64 |
| 3. 5^6 | 12. 2^3 | 22. 243 |
| 4. 2^2 | 13. 36 | 23. 81 |
| 5. 4^2 | 14. 49 | 24. 16 |
| 6. 2^5 | 15. 27 | 25. 125 |
| 7. 7^3 | 16. 100 | 26. 9 |
| 8. 8^4 | 17. 81 | 27. 16 |
| 9. 9^6 | 18. 8 | 28. 32 |
| | 19. 64 | |

Medium (drill)

- | | | |
|---------------------------|------------------------------|---------------------------|
| 1. base 5, index 2, = 25 | 8. base 3, index 3, = 27 | 15. base 4, index 2, = 16 |
| 2. base 2, index 4, = 16 | 9. base 6, index 3, = 216 | 16. base 4, index 3, = 64 |
| 3. base 3, index 2, = 9 | 10. base 4, index 4, = 256 | 17. 23 |
| 4. base 2, index 3, = 8 | 11. base 6, index 2, = 36 | 18. 25 |
| 5. base 7, index 2, = 49 | 12. base 7, index 3, = 343 | 19. 34 |
| 6. base 5, index 4, = 625 | 13. base 2, index 2, = 4 | 20. 25 |
| 7. base 3, index 4, = 81 | 14. base 7, index 4, = 2 401 | 21. 13 |
| | | 22. 80 |

Challenging

1. 256
2. $64 - 27 = 37$
3. $8 \times 9 = 72$

Extension

1. $3^4 = 9^2$
2. $n = 10$

Exit ticket

1. Level 1: 7^3
2. Level 2: 64
3. Level 3: $9 \times 16 = 144$

3.08 Square roots and cube roots (booklet p. 31)

WE DO examples (teacher)

1. Ex 1a: 7
2. Ex 1b: 12

3. Ex 2a: 20
4. Ex 2b: 9
5. Ex 3a: $\sqrt{144} = 12$
6. Ex 3b: 7 and 8

Easy (drill)

- | | | |
|--------|---------|---------|
| 1. 2 | 12. 1 | 23. 9 |
| 2. 6 | 13. 13 | 24. 144 |
| 3. 22 | 14. 3 | 25. 25 |
| 4. 14 | 15. 5 | 26. 81 |
| 5. 20 | 16. 19 | 27. 100 |
| 6. 17 | 17. 289 | 28. 256 |
| 7. 21 | 18. 400 | 29. 121 |
| 8. 15 | 19. 49 | 30. 169 |
| 9. 12 | 20. 36 | 31. 196 |
| 10. 24 | 21. 4 | 32. 225 |
| 11. 25 | 22. 1 | |

Medium (drill)

- | | | |
|--------|---------|---------------|
| 1. 2 | 9. 1000 | 17. 7 and 8 |
| 2. 729 | 10. 27 | 18. 7 and 8 |
| 3. 216 | 11. 3 | 19. 10 and 11 |
| 4. 125 | 12. 5 | 20. 12 and 13 |
| 5. 64 | 13. 8 | 21. 4 and 5 |
| 6. 9 | 14. 10 | 22. 12 and 13 |
| 7. 8 | 15. 4 | |
| 8. 343 | 16. 1 | |

Challenging

1. 9 and 10
2. 4 and 5
3. $\sqrt{100} = 10$; no ($6 + 8 = 14$)

Extension

1. 70 and 30
2. 7 cm

Exit ticket

1. Level 1: 8
2. Level 2: 5
3. Level 3: $11 - 10 = 1$

3.09 Prime and composite numbers (booklet p. 35)

WE DO examples (teacher)

1. Ex 1a: 1, 2, 3, 4, 6, 12; 1, 13
2. Ex 1b: neither
3. Ex 2a: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29
4. Ex 2b: No: $51 = 3 \times 17$
5. Ex 3a: 29, 31, 37, 41, 43, 47
6. Ex 3b: 46, 48, 49, 50, 51, 52, 54, 55, 56, 57, 58

Easy (drill)

- | | | |
|-------|------------------------------|-------------------------------|
| 1. C | 11. P | 20. 1, 31 |
| 2. C | 12. C | 21. 1, 41 |
| 3. C | 13. C | 22. 1, 2, 3, 4, 6, 12 |
| 4. C | 14. P | 23. 1, 5, 7, 35 |
| 5. C | 15. C | 24. 1, 3, 5, 9, 15, 45 |
| 6. C | 16. C | 25. 1, 2, 4, 11, 22, 44 |
| 7. C | 17. 1, 2, 3, 4, 6, 8, 12, 24 | 26. 1, 2, 23, 46 |
| 8. P | 18. 1, 13 | 27. 1, 3, 9 |
| 9. C | 19. 1, 5, 25 | 28. 1, 2, 3, 6, 7, 14, 21, 42 |
| 10. C | | |

Medium (drill)

- | | | |
|------|------|-------|
| 1. 5 | 3. 2 | 5. 12 |
| 2. 4 | 4. 4 | 6. 2 |

7. 8	13. 4	19. e.g. 3×33
8. 8	14. 4	20. e.g. 2×32
9. 2	15. 3	21. P
10. 4	16. 6	22. e.g. 2×64
11. 6	17. P	
12. 4	18. e.g. 2×46	

Challenging

- every other even number has 2 as a factor
- e.g. $5 + 31$, $7 + 29$, $13 + 23$, $17 + 19$
- 3,5; 5,7; 11,13; 17,19; 29,31; 41,43

Extension

- yes (not divisible by 2, 3, 5, 7, 11, 13)
- no: 13×17

Exit ticket

- Level 1: prime
- Level 2: 1, 2, 4, 7, 14, 28
- Level 3: 31, 37, 41, 43

3.10 Prime factors (booklet p. 39)

WE DO examples (teacher)

- Ex 1a: $2^2 \times 3 \times 5$
- Ex 1b: $2^3 \times 5^2$
- Ex 2a: $2^4 \times 3^2$
- Ex 2b: $2 \times 3^2 \times 5$
- Ex 3a: 2^8 , so $2^4 = 16$
- Ex 3b: $2^3 \times 3^6$, so $2 \times 3^2 = 18$

Easy (drill)

1. 2×13	9. 2×7	17. $2^2 \times 7$
2. 5×13	10. 5×7	18. $3^2 \times 5$
3. 3×13	11. 3×11	19. 3×19
4. 2×11	12. 2×2	20. $2^3 \times 3^2$
5. 7×7	13. $2^2 \times 5$	21. 3×5
6. 11×11	14. 2^3	22. 2×3^3
7. 3×3	15. $2 \times 3 \times 11$	23. $2^3 \times 3$
8. 7×13	16. 2×23	24. $2 \times 5 \times 7$

Medium (drill)

1. $2^3 \times 5 \times 7$	9. 19^2	17. 12
2. $2 \times 3^2 \times 17$	10. 10	18. 90
3. 3×113	11. 30	19. 4
4. 11×17	12. 40	20. 72
5. $2^2 \times 71$	13. 20	21. 60
6. 17×23	14. 2	22. 24
7. $2^7 \times 3$	15. 36	23. 120
8. 2×3^4	16. 18	24. 6
		25. 360

Challenging

- $2^8 \rightarrow 16$
- $2^2 \times 3^4 \rightarrow 18$
- $2^3 \times 3^6 \rightarrow 18$

Extension

- 5
- 12

Exit ticket

- Level 1: $2^2 \times 5$
- Level 2: $2^2 \times 3 \times 7$
- Level 3: $2^6 \times 3^2 \rightarrow 24$

3.11 Highest common factor (booklet p. 45)

WE DO examples (teacher)

- Ex 1a: 6
- Ex 1b: 4
- Ex 2a: 12
- Ex 2b: 17
- Ex 3a: 81
- Ex 3b: 36

Easy (drill)

1. 15	11. 7	21. 1, 2, 4, 8
2. 4	12. 6	22. 1, 2, 5, 10
3. 4	13. 5	23. 1, 5
4. 2	14. 3	24. 1, 3, 5, 15
5. 7	15. 6	25. 1, 5
6. 5	16. 8	26. 1, 2, 5, 10
7. 6	17. 1, 2, 4	27. 1, 2, 3, 6
8. 9	18. 1, 2, 4	28. 1, 2
9. 7	19. 1, 2, 4, 8	
10. 5	20. 1, 3	

Medium (drill)

1. 36	9. 16	17. 8
2. 9	10. 11	18. 2
3. 6	11. 4	19. 9
4. 8	12. 7	20. 10
5. 56	13. 7	21. 2
6. 10	14. 24	22. 5
7. 18	15. 60	
8. 18	16. 32	

Challenging

- $3/7$
- 8
- 15 cm

Extension

- 24
- 15

Exit ticket

- Level 1: 4
- Level 2: 14
- Level 3: 84

3.12 Lowest common multiple (booklet p. 49)

WE DO examples (teacher)

- Ex 1a: 12
- Ex 1b: 24
- Ex 2a: 90
- Ex 2b: 35
- Ex 3a: 400
- Ex 3b: 180

Easy (drill)

1. 72	11. 15	21. 21, 42, 63
2. 12	12. 12	22. 24, 48, 72
3. 56	13. 60	23. 10, 20, 30
4. 22	14. 18	24. 4, 8, 12
5. 9	15. 30	25. 8, 16, 24
6. 10	16. 42	26. 20, 40, 60
7. 6	17. 8, 16, 24	27. 21, 42, 63
8. 60	18. 15, 30, 45	28. 6, 12, 18
9. 10	19. 6, 12, 18	
10. 56	20. 42, 84, 126	

Medium (drill)

- | | | | |
|--------|---------|---------|---|
| 1. 20 | 9. 105 | 17. 24 | 2. 24 |
| 2. 45 | 10. 56 | 18. 140 | 3. 4 packs of hot dogs, 5 packs of buns (40 each) |
| 3. 66 | 11. 399 | 19. 360 | |
| 4. 255 | 12. 60 | 20. 60 | |
| 5. 132 | 13. 312 | 21. 20 | |
| 6. 14 | 14. 12 | 22. 10 | |
| 7. 40 | 15. 221 | | |
| 8. 357 | 16. 90 | | |

Challenging

1. 7:00 am (60 minutes)

Extension

1. 72
2. 60 seconds

Exit ticket

1. Level 1: 12
2. Level 2: 48
3. Level 3: 84